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2023 International Consensus Meeting on musculoskeletal infection: Summary from the treatment workgroup and consensus on treatment in preclinical models

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Abstract

In vitro and in vivo studies are critical for the preclinical efficacy assessment of novel therapies targeting musculoskeletal infections (MSKI). Many preclinical models have been developed and applied as a prelude to evaluating safety and efficacy in human clinical trials. In performing these studies, there is both a requirement for a robust assessment of efficacy, as well as a parallel responsibility to consider the burden on experimental animals used in such studies. Since MSKI is a broad term encompassing infections varying in pathogen, anatomical location, and implants used, there are also a wide range of animal models described modeling these disparate infections. Although some of these variations are required to adequately evaluate specific interventions, there would be enormous value in creating a unified and standardized criteria to animal testing in the treatment of MSKI. The Treatment Workgroup of the 2023 International Consensus Meeting on Musculoskeletal Infection was responsible for questions related to preclinical models for treatment of MSKI. The main objective was to review the literature related to priority questions and estimate consensus opinion after voting. This document presents that process and results for preclinical models related to (1) animal model considerations, (2) outcome measurements, and (3) imaging.

KEYWORDS

antibiotic, biomarkers, fracture, imaging, infection

1 | INTRODUCTION

Musculoskeletal infections (MSKI) include infections of the bone, muscles, or joints and encompass periprosthetic joint infection (PJI), fracture-related infection (FRI), septic arthritis, vertebral osteomyelitis as well as native osteomyelitis. Even with optimal and appropriate preventative measures, MSKI occurs at a rate of 1%–2% for prosthetic joint replacements and 5%–30% for traumatic injuries such as open fractures.[1] Infection rates are even higher for military cohorts and may compromise 63% of nonbattle related injuries and nearly 90% of injuries among deployed service members.[2,3]

Once established, MSKI are a devastating complication, requiring (re)hospitalization, repeated surgery, increased outpatient visits, and lead to bone defects requiring prolonged reconstruction, functional impairments, and reduced quality of life[4,5] The treatment of established MSKI is generally centered around surgical debridement and antibiotic therapy. Current treatment options are built around surgical interventions, being either implant retention or removal, or one or two-stage exchange, with the presence and presumed maturity of the biofilm being a major deciding factor. Despite the availability of algorithms aiding in selecting the optimal treatment approach, treatment success rates for MSKI vary widely and are often poorest in cases where such algorithms cannot be followed. The consequences of failed therapy include infection recurrence, progression to chronic infection, and potentially the creation of large bone defects due to excision of infected tissues. These factors all drive substantially increased patient morbidity and healthcare costs.

The clinical and economic implications of MSKI have driven multiple lines of research into novel antimicrobials, antimicrobial-laden bone void fillers and wound management strategies to increase treatment success rates. Research and development exploring and evaluating innovations in the treatment of MSKI require preclinical models capable of mimicking and monitoring the development and progression of infection, and its response to therapy. Although MSKI is regularly studied in animal models, there seems to be little if any agreement on which model to use in preclinical testing, with individual institutions often establishing and using their own model with little transfer across the field. The lack of consistency in methodology can contribute to difficulty in comparing results between studies, failure to use best practices in all aspects of study execution (e.g., animal welfare, infection induction and treatment, outcome measures etc.). In 2023, an international consensus meeting (ICM) focused on preclinical models of MSKI was held at the Orthopaedic Research Society (ORS) Annual Meeting to establish expert recommendations for the use and evaluation of preclinical infection research. The workgroup was established to focus on issues related to animal models studying infection treatment. Led by a panel of three co-chairs, the working group identified key questions, assigned to a wider panel of delegates and co-authors, and managed the literature review, preparation of summary statements, and assembled the vote on recommendations. This consensus article

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provides these responses in addition to commentary on future research goals arising from this initiative.

2 | METHODOLOGY

The main objective of ICM 2023 was to collect and assemble the best available data on preclinical models of MSKI with input from experts around the world to formulate a consensus document (https://ors-org.s3.amazonaws.com/wp-content/uploads/2023/06/ORS-MSKI-RIG-ICM-Final.pdf). Preclinical models are used to evaluate treatment outcomes for novel therapeutics or protocols and thus directly impacts patients impacted by implant-related, bone and joint infections (MSKI), including surgical site infections, PJI, and FRI. The first step, led by ICM 2023 cochairs, Drs. Edward Schwarz, Fintan Moriarty, Antonia Chen, Tom Schaer, Kordo Saeed, and Noreen Hickok were to identify and recruit an international panel of MSKI experts. The experts served as delegates tasked to identify the common practices and challenges related to preclinical models of MSKI, then seek consensus on those issues using Delphi methodology[6] to achieve "general agreement" in response to questions.

Delegates with well-founded expertise in the field of MSKI were invited and generated questions addressing clinical issues related to MSKI. Questions were categorized into research areas, each addressed by separate work groups, including a workgroup to address issues related to treatment. Over 24 months, each question was assigned to two or more delegates for response preparation. Response preparation consisted of a literature review, data summary, and an independent narrative response written from the perspective and experience of each authoring delegate. Responses were reviewed by facilitators and compiled in a document. The compiled responses were then edited by all authors to an agreed response to be distributed to all delegates for review and comment. The authoring delegates refined their responses based on the comments in preparation for discussion and voting at the in-person consensus meeting held on February 10, 2023, in Dallas, TX, USA.

The controversial questions and responses were further discussed and edited for online voting. The delegates then voted to (1) agree; (2) disagree; or (3) abstain, on each, following Delphi methodology[7] and the voting results were rated as (a) Simple majority (50.1%–59%); No Consensus; (b) Majority (60%–65%); Weak Consensus; (c) Super Majority (66%–99%); Strong Consensus; and (d) Unanimous (100%): Unanimous Consensus. Among the work groups, there was one made up of the 33 authors in this consensus document dedicated to treatment. This group consisted of infection experts from biomedical engineering, immunology, infectious diseases, microbiology, and orthopaedic surgery, among other disciplines. The treatment workgroup was tasked with discussion and edits of the 12 treatment-related questions that were considered relevant for preclinical research. The Workgroup emphasizes that consensus was reached without compulsion, undue influential power, inability to comprehend another course of action, or impatience with the process of discussion and voting. Below is a summary of the 12 treatment-related questions, responses, and/or recommendations, with HTML links to a downloadable PDFs for each question, response, consensus, and postmeeting rationale (https://ors-org.s3.amazonaws.com/wp-content/uploads/2023/06/ORS-MSKI-RIG-ICM-Final.pdf) and the ICM voting results (https://ors-org.s3.amazonaws.com/wp-content/uploads/2023/06/ORS-MSKI-ICM-Vote.pdf) are available on the ORS website.

3 | RESULTS

The Treatment Workgroup's responses to the 12 questions are summarized in Table 1, which covers questions related to infection model and inoculation, assessment of treatment efficacy, and complementary imaging methods for evaluation of tissue response. The strength of evidence assigned to each response is based on the quality of studies and scientific validity of data and agreed upon by members of the workgroup. There were equal numbers of Strong and Moderate Level evidence for treatment-based questions, with two answers having responses based on low or limited evidence from reported literature. Here we present an interpretive discussion of the responses, grouped according to three topics of animal model considerations, outcome measurements, and imaging. The readership is encouraged to access the full rationale for each question in the ICM 2023 document (https://ors-org.s3.amazonaws.com/wp-content/uploads/2023/06/ORS-MSKI-RIG-ICM-Final.pdf) to gain a deeper understanding of the available data.

3.1 | Animal model considerations

Six questions were related to details on the animal model considerations, including duration, availability of models with specific treatment modalities (surgical and antibiotic-related), bacterial inoculum, and antibiotic selection.

The first question related to the timeline for observation posttreatment to determine efficacy of treatment. The answer provided by expert review stated "No," and a super majority agreed that there was not a standard timeline to best evaluate the resolution of infection in animal models of MSKI. In response to Question 16, delegates voted 88.9% in agreement, 5.6% in disagreement, and 5.6% abstained. The answer was motivated by the fact that in any study, the results will depend on multiple variables: infectious agent, the extent of bone infection, anatomic location, and/or the animal species used. This situation also complicates human clinical scenarios of infection. The recommendation called for using clinical paradigms to guide model selection, driven by considerations of the clinical scenario being modeled, inoculum type, bone remodeling/response, and current recommended practices; there are many scenarios that accompany MSKI. While true, it is also fair to say that specific injuries or pathogens are more common in MSKI. From these more prevalent scenarios, a timeline may be established on a case-specific basis. Still, overall, clinical guidelines for managing infection lack a clear

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TABLE 1 Treatment workgroup-related questions, responses, and level of agreement by ICM delegates.

Question 16	**Is there a standard timeline or duration of treatment to best evaluate resolution of infection in animal models of MSKI?**
Answer	No
Rationale	Timelines for studying infection resolution in animal models of osteomyelitis, fracture-related infection, and PJI are highly dependent on the model, i.e., infectious agent, the extent of bone infection, anatomical location, and the species of animal used. Even in human clinical scenarios, these timelines are highly variable from patient to patient under similar circumstances. Currently, there is no definitive evidence to indicate the optimal timelines of infection treatment for in vivo models of osteomyelitis, facture-related infection, and PJI. We recommend using existing clinical paradigms to guide model selection and study timelines. This is guided by four considerations: clinical scenario being modeled, inoculum type, bone remodeling/response, and current recommended practices.
Strength	Moderate
Vote	Agree: 89, Disagree: 6, Abstain: 5
Result	Super Majority, Strong Consensus
Question 17	**Does radiography score correlate to infection and treatment efficacy for bone?**
Answer	Yes
Rationale	Radiographic scoring strongly correlates to bone infection when X-ray imaging is performed 1–2 weeks or greater following infection. Radiographic scoring also strongly correlates to treatment efficacy for bone infection when X-ray imaging is performed following substantial differences in infection burden between groups over a duration of approximately 3 weeks or greater. Limitations in radiographic scoring include the inability to detect differences in early infection and quantify early treatment efficacy. Limitations in current literature assessment include variability in radiographic scoring methodology and animal model used.
Strength	Moderate
Vote	Agree: 82, Disagree: 8, Abstain: 10
Result	Super Majority, Strong Consensus
Question 18	**Are there any imaging techniques available that can effectively indicate the degree of infection or monitor the advancement of the disease in animal models of MSKI?**
Answer	Yes
Rationale	Yes There are imaging techniques available that can effectively indicate the degree of infection or monitor the advancement of the disease in animal models of MSKI. X-ray can provide basic assessment of trabecular and cortical bone structure changes over longitudinal assessment during the development of MSKI. Micro-CT is valuable for revealing microarchitecture of periosteal reaction, medullary low-attenuation areas, trabecular coarsening, and focal cortical erosions with high precision in small animal models. However, both X-ray and micro-CT are limited in detecting features of disease progression often days to a week after the initiation of infection and treatment. MRI is more sensitive to the presence of soft-tissue abnormalities and acute osteomyelitis than X-ray and CT due to its in-depth determination of tissue swelling or fluid formation, formation of ducti, evaluation of soft tissue involvement in infection, and assessment of angiogenesis. PET/CT shows promise in detecting infection boundaries and likely has the potential to define the scope of lesions more accurately. Additionally, PET/CT can calculate blood flow through a specific area of interest in real time. Finally, some novel imaging techniques, such as nano-CT, are expected to improve the detection of MKSI advancement in in vivo models.
Strength	Strong
Vote	Agree: 92, Disagree: 3, Abstain: 5
Result	Super Majority, Strong Consensus
Question 19	**Are IVIS using fluorescence or luminescence complementary to other methods such as culture or PCR?**
Answer	Yes
Rationale	Yes We conclude that IVIS provides complementary information on the bacterial burden to culture and PCR, and its usage is particularly valuable in small animal models for in vivo longitudinal assessments. Limitations of IVIS, compared to CFU analysis or PCR, include less reliable detection and quantification of infection at low bacterial concentrations and infections localized deep underneath tissue. Furthermore, since the production of bioluminescent signals relies on metabolically active bacteria, data obtained by IVIS are less reliable when metabolic activity is decreased (such as when biofilm is present). This should be taken into consideration in the design of MSKI studies.
Strength	Strong
Vote	Agree: 93, Disagree: 4, Abstain: 3

(Continues)

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TABLE 1 (Continued)

Result	Super Majority, Strong Consensus
Question 20	**Can techniques be employed to precisely evaluate the formation of biofilms on implants or infected bone in in vivo models of osteomyelitis?**
Answer	Yes
Rationale	Yes Biofilm formation in vivo differs structurally from in vitro biofilm, as the latter lacks the mushroom-like structure and is smaller in size 1. In vivo models of implant-related bone infection demonstrate complex interactions between the host, the implants and microbes, leading to milieu bacterial adhesion.Various techniques are available to assess the formation of biofilms on implants or infected bone in vivo models of osteomyelitis. These include microscopic analysis, biomolecular assays, BLI, X-ray imaging, and histological analysis. Microscopic analysis, such as SEM, FE-SEM, and CLSM, allows for detailed visualization and characterization of biofilm structure and composition. Biomolecular assays, like ELISAs, enable the detection and quantification of specific molecules associated with biofilm formation. BLI provides real-time monitoring of biofilm formation using genetically engineered bacteria that emit light under specific conditions. X-ray imaging, specifically micro-CT, allows for visualization of biofilm distribution on implants or infected bone. The histological analysis enables visualization of infection extent and biofilm formation in surrounding tissue. The selection of technique(s) depends on the research question and the type of implant or infected bone being studied. Combining multiple techniques can provide a more comprehensive understanding of biofilm formation in vivo.
Strength	Strong
Vote	Agree: 89, Disagree: 1, Abstain: 10
Result	Super Majority, Strong Consensus
Question 21	**Is there an animal model representative of DAIR?**
Answer	Yes
Rationale	Much has been published on PJI and DAIR models in rodents and rabbits; however, little information is available on DAIR models for other animals. Few PJI models contain implants materially similar to those used clinically. Further, the amount and strain of inoculation vastly differ between studies, which makes a determination of an appropriate DAIR intervention timepoint difficult. Additional work should include implant resemblance, inoculation technique, and time to DAIR intervention as considerations.
Strength	Moderate
Vote	Agree: 83, Disagree: 6, Abstain: 11
Result	Super Majority, Strong Consensus
Question 22	**Is there a single predominant combination of bacterial species in humans that should be studied in animal models of polymicrobial MSKI?**
Answer	No
Rationale	Polymicrobial infections do not have a particular combination of pathogens and are rare in the context of MSKIs. While no particular strain is responsible for such infections, gram-negative bacteria may be more common in polymicrobial infections. The incidence of polymicrobial infections depends on the type and location of the infection (osteomyelitis, fracture infections, spondylodiscitis, etc.), and are caused by various strains of bacteria. These strains include Gram-negative bacilli, nonepidermidis coagulase-negative Staphylococci, *Staphylococcus epidermidis*, and *Staphylococcus aureus*. Animal models of polymicrobial infections are limited in literature and have difficulties like determining the optimal bacterial inoculum. Therefore, polymicrobial infections in animal models should be aimed at answering specific research questions (synergism, interactions, etc.), and may contain gram-negative strains.
Strength	Low
Vote	Agree: 85, Disagree: 7, Abstain: 8
Result	Super Majority, Strong Consensus
Question 23	**Are there immunological plasma biomarkers that are useful to measure infection or treatment effects in rat models of MSKI?**
Answer	Yes
Rationale	There are markers which have demonstrated efficacy in longitudinal rat inflammation and infection studies. Haptoglobin, α2M, fibrinogen, TNF-α, and CRP have demonstrated correlation with infection and inflammation in rats shortly (~1–7 days) after infection, with or without injury. Data on these markers as indicators of treatment effect are limited, with a few studies reporting utility of haptoglobin. Due to varying pathogeneses of infectious agents and levels of inflammation, there is likely not a single superior marker for tracking infection or treatment efficacy in rats. Recent studies analyzing multiple plasma markers and cell types have begun to enable a more complete understanding of the immune response to infection.
Strength	Moderate

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TABLE 1 (Continued)

Vote	Agree: 86, Disagree: 6, Abstain: 8
Result	Super Majority, Strong Consensus
Question 24	**Can all SOC antibiotics included in the clinical treatment guidelines for MSKI be used in animal models?**
Answer	Yes
Rationale	All SOC antibiotics can be used in animal models of MSKI. The optimal choice of antibiotic therapy relies on a variety of criteria including the animal involved, the presence of a foreign body, causative agent, the presence of a biofilm, and the aim of the study. All first-line antibiotics have been utilized in animal models of MSKI, which frequently continues for multiple weeks or until the causal agent is identified. This demonstrates that antibiotic treatment using SOC drugs is feasible in animals.
Strength	Strong
Vote	Agree: 90, Disagree: 3, Abstain: 7
Result	Super Majority, Strong Consensus
Question 25	**Consolidated with question 24**
Question 26	**Is a statistically significant reduction in bacterial burden clinically significant if infection remains after treatment in animal models of infection?**
Answer	Yes
Rationale	The literature linking changes in CFU counts in animal models with clinically significant changes in infection is weak and circumstantial. Despite these weaknesses, there does appear to be some evidence that bacterial loads generated in current in vivo models does replicate findings from clinical studies in the chronic wound literature in humans. Additionally, thresholds below 103–104 CFU, which would correlate in most studies to a 2-log reduction in bacterial load would be below the threshold that has been used clinically to roughly define appropriate wound healing and skin colonization versus active infection in chronic wound healing studies in humans. This would probably be a minimum reasonable threshold to use for in vivo studies.
Strength	Limited
Vote	Agree: 69, Disagree: 18, Abstain: 13
Result	Super Majority, Strong Consensus
Question 27	**After using an antimicrobial-loaded biomaterial in an animal study, can you prevent false negative culture results due to antimicrobial carryover during sample processing in the lab?**
Answer	Yes
Rationale	False negative cultures occur when study antibiotics or antiseptics kill bacteria or inhibit culture after the time of sampling. In the case of antibiotics, super-MIC antibiotic concentrations accompany bacteria into culture plates. To this end, preventing false negatives requires one of several techniques to either dilute or remove antibiotics such that antibiotic concentrations are below MIC values when culturing, e.g., pelleting/washing or neutralization in combination with good laboratory practices and controls.
Strength	Strong
Vote	Agree: 90, Disagree: 4, Abstain: 6
Result	Super Majority, Strong Consensus
Question 28	**Should treatment of fracture and/or implant-related infection always include debridement in animal models?**
Answer	Yes
Rationale	Debridement should always be considered as part of the SOC for treating fracture and/or implant-related infections in animal models.
Strength	Moderate
Vote	Agree: 71, Disagree: 18, Abstain: 11
Result	Super Majority, Strong Consensus

Abbreviations: α2M, alpha2-macroglobulin; BLI, bioluminescence imaging; CFU, colony forming unit; CLSM, confocal laser scanning microscopy; CRP, c-reactive protein; CT, computed tomography; DAIR, debridement, antibiotics, and implant retention; ELISA, enzyme-linked immunosorbent assay; FE-SEM, field-emission scanning electron microscopy; ICM, international consensus meeting; IVIS, in vivo imaging systems; MIC, minimum inhibitory concentration; MRI, magnetic resonance imaging; MSKI, musculoskeletal infections; PCR, polymerase chain reactio; PET, positron emission tomography; PJI, periprosthetic joint infection; SEM, scanning electron microscopy; SOC, standard of care; TNF-α, tumor necrosis factor-alpha.

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consensus. Thus, it is challenging to offer a standard timeline in an animal model, especially considering that human cases of chronic infection may extend from weeks to months. The creation of a definitive timeline for infection resolution in animal models of MSKI must start with specific considerations related to the condition of interest and may best be performed for the most common pathogens for defined infection types.

In response to Question 21 about whether animal models of drbridement, antibiotics, and implant retention (DAIR) are available. The recommendation response was, "Yes, there have been several animal models for DAIR," with no perceived controversy. Delegates voted 83.3% in agreement, 5.6% in disagreement, and 11.1% abstained. Much has been published on PJI and DAIR models in rodents and rabbits. However, little information is available on DAIR models for other animals.[8–12] While there is no ideal model or one that very closely replicates clinical scenarios, moderate evidence supports that modest DAIR models exist. Few PJI models contain implants similar to those used clinically. Further, there is a wide variety between studies in the amount and strain of inoculationmakes determination of an appropriate DAIR intervention timepoint difficult. Future work is necessary to develop more clinically relevant models that allow for infection to develop, DAIR to be performed in a meaningful manner, and allow timeframes of healing to occur with outcome measures modeling clinical practice, for example, white blood cell counts, fever, swelling, erythrocyte sedimentation rate, cytokine analysis, and other metrics as deemed relevant to the clinical scenario being modeled. Additional work should include implant functionality, inoculation technique, and time to DAIR intervention as considerations.

Question 22 addresses polymicrobial infections in animal models of MSKIs and how these models should be designed. A super majority, 85% agreed with the response of "No" to the question of whether there is a single predominant combination of bacterial species in humans that should be studied in animal models of polymicrobial MSKI, 7% disagreed, and 8% abstained. Depending on the etiology (osteomyelitis, PJI, FRI, spondylodiscitis, and infected nonunions) and local epidemiology, the prevalence of polymicrobial infections in humans ranges from about 10% to 25% associated with a wide range of pathogens with no particular combination dominating.[13–15] About 65% of these infection cases are caused by Gram-negative bacilli, followed by nonepidermidis coagulase-negative Staphylococci, *Staphylococcus epidermidis*, and *Staphylococcus aureus*. In polymicrobial infections, a complex environment may be formed, in which microbiological synergistic interactions exist between microorganisms. This microbial synergism, in line with the need for broad-spectrum antimicrobial therapy, may result in a more challenging treatment, but also adverse outcomes for patients suffering from MSKIs.[16,17] Limited literature exists on polymicrobial infections of the musculoskeletal system in animal models. Available models include polymicrobial infections induced by *S. aureus* in combination with *Pseudomonas aeruginosa*, while one model instead examines co-infection with *Escherichia coli*.[18–21] These models have encountered challenges in establishing appropriate bacterial inoculum concentrations to induce infection without compromising animal welfare. Nevertheless, no serious complications or high rates of animal exclusion have been reported in these studies.[18–21] Therefore, polymicrobial infections in animal models should aim to answer specific research questions (synergism, interactions, etc.), and may contain Gram-negative strains. In conclusion, animal models for polymicrobial MSKIs should be limited to specific research questions such as pathogen interactions or specific treatment approaches. If such a model is planned, it may contain at least one Gram-negative pathogen, consistent with the microbial etiology in MSKIs.

Question 24 addresses the use of standard-of-care (SOC) antibiotics in animal models of MSKI. Specifically, the question asked, "Can all standard-of-care antibiotics included in the clinical treatment guidelines for musculoskeletal infection be used in animal models?" Delegates voted 90% stating, "Yes," the SOC antibiotics used in clinical treatments can be included in the clinical guidelines for MSKI in animal models, 3% voted in disagreement, and 7% abstained. The optimal choice of antibiotic therapy relies on a variety of criteria including the animal involved, presence of a foreign body, causative agent, presence of a biofilm, and the aim of the study. In human patients, treatment is often started empirically with broad-spectrum antibiotics and switched to a more targeted approach when the causative pathogen is known. Clinical guidelines are developed to advise the optimal treatment strategy based on the pathogen, antibiotic resistance patterns, and the presence of foreign materials (e.g., prosthesis or osteosynthesis material). Understandably, the preclinical studies that were found to address question 24 focused on the most frequently encountered bacteria in MSKIs (e.g., Staphylococcal species, Enterobacteriaceae, and *P. aeruginosa*), and so antibiotic therapies described in the literature are most often targeting these pathogens, with more unusual pathogens rarely, if ever, studied. The systematic review of literature confirmed that all antibiotics, recommended by the clinical guidelines to treat infections with the most common pathogens, have been administered previously in animal models. Data was collected on the administration route, the total daily dose, and the duration of antibiotic administration. However, whether the use of using such a large variety of antimicrobials in preclinical studies is favorable over a more standardized approach was not assessed during the consensus meeting. The need for using a specific antibiotic is obvious with certain research questions, like evaluating bioavailability or biofilm activity of a particular drug. When the study does not directly aim to evaluate the systemic antibiotic itself, it might be preferable to pursue a more standardized approach and use only first-line antibiotics of the clinical guidelines. This would reduce the range of antibiotics tested and improve consistency between different studies and limit the effect of a difference in systemic antibiotic regimens on the study outcome. The development of a more regulated approach would require additional information on translating antibiotic effects from animals to humans, optimal dosing strategies, and the safety and tolerability of different treatment strategies.

Fracture and/or implant-related infection has a high morbidity burden in orthopaedic patients and a prompt diagnosis is an important

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step in the treatment process. Question 28 asked, "Should treatment of fracture and/or implant-related infections always include debridement in animal models?." The delegate votes were 71% in agreement, 18% disagreement, and 11% abstained for the recommendation of "yes." Diagnostic criteria suggested by Metsemakers et al.[22] and Govaert et al.[23] concluded presence of sinus, fistula, or wound breakdown (with communication to the bone or implant) and/or purulent drainage from the wound or presence of pus during surgery and these are pathognomonic for the diagnosis of FRI.[22,23] Any other clinical sign of infection, apart from the aforementioned, are considered suggestive and should prompt further investigation from physician or surgeon. If there is presence of any of the confirmatory criteria, debridement should be considered as part of SOC since it helps remove necrotic tissue and biofilm matrix, reduces bacterial count, and prevents progression of biofilm formation. The consensus voting results for the question, "Should treatment of fracture and/or implant-related infection always include debridement in animal models?" was one of the lowest agreements in the process at 70%. Although speculative, reasons behind this may be due to the practical issues of repeated surgery and the technical expertise required not being available in many basic research settings. However, considering the central role played by debridement in the clinic, it seems logical to include it in animal studies whenever appropriate or possible. Any treatment strategy that successfully treats MSKI in the absence of debridement would of course be a powerful estimation of efficacy.

3.2 | Outcome measurements

Question 20 investigates if techniques be employed to precisely evaluate the formation of biofilms on implants or infected bone in in vivo models of osteomyelitis. Delegates voted 89% in agreement, 1% disagreed that "Yes, techniques can be employed to precisely evaluate the formation of biofilms on implants or infected bone in in vivo models of osteomyelitis" and 10% abstained. The high number of abstentions indicates that many may have been confused about the meaning of the answer and its relation to preclinical models of treatment or did not feel qualified to answer the question. Microscopy (histology) is routinely used in clinics for diagnosing and evaluating infection of the bone and can be utilized as a viable approach for probing biofilms formation on implants in labs with specialized equipment. While these are routine, they are repeatedly modified and there is no standard technique for imaging and analysis at a high resolution. Scanning electron microscopy is a standard technique used for studying the surface morphology; however, the technique cannot examine qualitative and quantitative information on the biofilm structure at high resolutions.[24] It is generally implicit that biofilm on implants should be detected, but there may be disagreement on the level of precision current techniques offer. Advancements in biofilm detection and measurement techniques are needed to improve the rigor of preclinical models evaluating treatments.

Question 23 assesses whether any biomarkers may be useful as indicators of infection or treatment effects in rat models of MSKI. Delegates voted 86.1% in agreement, 5.6% in disagreement, and 8.3% abstained for the response of "Yes" markers demonstrating efficacy are available. The high number of abstained voters may correlate with the varying pathogeneses of infectious agents and levels of inflammation making broad statements challenging. Clinically, blood markers are used for diagnoses of infection in humans (e.g., elevated white blood cell counts, C-reactive protein [CRP]).[25] Notably, biomarkers useful in humans are not necessarily effective measures in rats (or other species) due to interspecies variability in levels of response and baseline levels. This warrants investigation of species-specific markers which effectively elucidate infection and/or treatment effects. Plasma biomarkers participate in complex signaling pathways, and their expression changes over time following infection and/or inflammation. While haptoglobin, alpha$_2$-macroglobulin, fibrinogen, tumor necrosis factor-alpha, and CRP have demonstrated correlation with infection and inflammation in rats shortly (~1–7 days) after infection, with or without injury, data on these markers as indicators of treatment effect are limited.[26] The lack of unanimous consensus may be due to this discrepancy. Further, as different pathogens modulate the immune response through various mechanisms which may alter bacterial load and mask treatment effect,[27] more data is needed to confirm each biomarker's effectiveness in distinguishing between infection models, and between treated and untreated groups within an infection model. Methods for quantifying these biomarkers include electrophoresis, enzyme-linked immunosorbent assays (ELISAs), and multiplex assays, but their use is inconsistent across studies, which makes interpreting results challenging. Electrophoresis and gel staining can distinguish multiple molecules by size from the same sample. ELISAs are highly accurate and sensitive but are generally designed for singular proteins or macromolecules, requiring large volumes of blood to quantify multiple markers. Multiple markers can be analyzed from the same sample simultaneously in bead-based multiplex arrays, but these require specialized instrumentation and can be less sensitive than single-factor ELISAs.[28] In some studies, statistical analyses were not performed on both blood markers and traditional outcome measures (i.e., colony forming unit [CFU] counts). In other studies, blood markers and traditional outcome measures were statistically analyzed, but effects of infection and/or treatment were not detected. A standardized, multiplex assay may be the most effective measure of immune response across models, though such a panel remains to be provided and represents an important future research target.

Question 26 addresses whether a statistically significant decrease in bacterial burden in animal models of infection is clinically significant. This is a challenging question because it is difficult to directly translate findings in an animal model to the human setting. The expert panel provided an answer of "Yes" although it was a qualified statement with limited strength of recommendation. The literature available to answer this question is mostly circumstantial and may not be directly clinically applicable. As a result, this was a controversial question among the delegates with only 69% in agreement, 18% disagreement, and 12.5% abstaining. Typical animal models use a 2- to 3-$\log_{10}$ reduction in bacterial burden with

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thresholds below 10^3–10^4 CFU to define "successful" treatment of underlying infection in experimental settings, and the thresholds of bacterial burden necessary to generate infection appear to be in a similar range.[29–33] Unfortunately, whether these findings directly translate to human infection is not clear. Some of the controversy lies in the lack of data regarding bacterial burden in human infection as it resolves with treatment. For instance, the thresholds for infection eradication in clinical osteomyelitis are not known due to the difficulty and lack of precise analysis at the infection site during treatment. As a result, microbial cultures are relied upon for this diagnosis, but are unreliable for several previously described reasons. Most of the literature in humans where bacterial burden was carefully characterized derives from the chronic wound literature, which may not directly translate to MSKI.[29,34–43] Nevertheless, in human wound healing, minimum intrawound bacterial loads of $\geq 10^4$ CFU/g tissue were needed to cause infection of complex extremity wounds and postsurgical wounds, and this also appeared to be the approximate threshold for successful skin grafting, which is consistent with bacterial burdens used in clinical animal models.[41–43] Further studies on longitudinally assessing bacterial burden during the healing process of skeletal infections would be valuable.

In response to Question 27, related to preventing false-negative culture due to antimicrobial carryover, delegates voted 90.3% in agreement, 4.2% in disagreement, and 5.6% abstained. The recommendation response was, "Yes, false negative cultures occur when study antibiotics or antiseptics kill bacteria or inhibit culture after the time of sampling," with no perceived controversy. Various methods exist to mitigate false negative culture results, primarily those that dilute the antimicrobial to a level where they are below the minimum inhibitory concentration and will not result in residual kill against bacteria. Future techniques could be improved by regulatory agencies or standardization organizations, for example, ISO or ATSM, establishing standards that apply to a broad range of culture techniques with specific intent to mitigate false culture negative potential.

3.3 | Imaging

The majority (81.9%) of the voting delegates of the ORS ICM agreed with the recommendation in question 17 that radiographic (X-ray) scoring correlates to infection and treatment efficacy for bone. However, 8.3% disagreed with this recommendation. Some of this disagreement could be due to limitations in X-ray detecting early and subtle changes with early infection or early treatment. Furthermore, when X-ray scoring is utilized, there is variability across studies in terms of X-ray scoring methodology, animal model, and/or therapeutic assessed. These limitations are reflected in our moderate strength of recommendation and may have led to some disagreement across some of the ICM delegates. While X-ray scoring can be a valuable tool for assessing bone infection and treatment of bone infection, developing and adopting consensus recommendations for X-ray scoring methodology for respective animal models with bone infection could enhance the generalizability of X-ray scoring across studies and increase agreement across ICM delegates.

Questions 18 and 19 both involve imaging techniques such as in vivo imaging systems (IVIS) and whether they can effectively monitor the degree of infection. Delegates for question 18 voted 92% in agreement, 3% disagreed, and 5% abstained. Both questions show the strongest level of agreement among the questions. Low disagreement confirms that there is little controversy about the advantages of imaging techniques. Question 18 specifically targets whether imaging techniques are available that can effectively indicate the degree of infection or monitor the advancement of the disease in animal models of MSKI. The majority of voters answered yes and referenced X-rays as a basic assessment of trabecular and cortical bone structure changes over longitudinal assessment during the development of MSKI. The advantages and disadvantages of microcomputed tomography (CT) and magnetic resonance imaging (MRI) were also explored. Studies show CT as a valuable imaging technique for revealing microarchitecture of periosteal reaction, medullary low-attenuation areas, trabecular coarsening, and focal cortical erosions with high precision in small animal models. However, both X-ray and micro-CT are limited in detecting features of disease progression often days to a week after the initiation of infection and treatment. MRI is more sensitive to the presence of soft-tissue abnormalities and acute osteomyelitis than X-ray and CT, due to its in-depth determination of tissue swelling or fluid formation, formation of ducti, evaluation of soft tissue involvement in infection, and assessment of angiogenesis. Positron emission tomography (PET)/CT shows promise in detecting infection boundaries and likely has the potential to define the scope of lesions more accurately. Additionally, PET/CT can calculate blood flow through a specific area of interest in real time. Finally, some novel imaging techniques, such as nano-CT, are expected to improve the detection of MKSI advancement in in vivo models. Related to IVIS using fluorescence or luminescence, the majority (93%) of delegates of the ORS ICM agreed with the recommendation that IVIS imaging methods are complementary to other methods such as culture or polymerase chain reaction (PCR). This is the highest level of agreement that IVIS provides complementary information to culture or PCR regarding bacterial burden. However, 4% of voters disagreed with this recommendation and 3% abstained. Some of this disagreement could be due to limitations in IVIS imaging, such as quantification of infection at low bacterial concentrations and infections localized deep underneath tissue. As IVIS technology continues to advance, such as the development of improved bioluminescent or fluorescent reporters, the limitations of IVIS may be further diminished.

4 | DISCUSSION

The treatment working group of the ORS ICM tackled important issues in the treatment of MSKI from different viewpoints. For animal model considerations, a clear variety of models exist, and

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that all front-line antibiotics may be readily administered to animals to closely replicate clinical treatment options. This paves the way for alignment on systemic antibiotic controls across animal models, which is not yet observed, but may be possible in future. The absence of a standard timeline, which may reflect the need to evaluate preclinical studies in similar contexts to clinical counterparts, does not have an obvious solution in the near term. However, here it may be possible to align protocols to aid cross comparison of studies, but at the present time, it remains unclear how to achieve this or convince research groups to deviate from protocols that are successful in the past to align with new protocols. There were higher levels of disagreement or abstentions for the recommendation that debridement always be included in fracture-related infection models, which may be dependent on the specific type of evaluation and the timing of intervention. The number of abstentions for questions related to DAIR models and polymicrobial models may reflect the lack of research on these topics and unfamiliarity of experts with their use. More research is needed to refine these models, and most recommendations agree that these should be representative of clinical scenarios, while at the same time, being perhaps mostly appropriate for certain research questions.

There were high levels of agreement for consistency in outcome measures such as taking measures to prevent false-negative cultures due to antimicrobial carryover and methods to evaluate biofilm on implants. Such practices are already described and encouraged by journals. The availability of biomarkers in preclinical models to measure treatment effects also had low levels of disagreement among delegates, confirming their role as complementary measures. Next steps for this topic include broadening to other species and nominating precise biomarker assays for optimal monitoring of outcomes including response to therapy. There were high levels of disagreement about the statement that a statistically significant reduction of bacterial burden is clinically significant, which may be related to the high risk of rebound infection if bacteria and biofilm are not completely eradicated at the site of injury. This topic is a good example of a topic where consensus opinion is a good indicator of prevailing thoughts on a topic, with no objective measure being available. Therefore, subjective, expert-led opinion is a useful tool to guide the field. The lack of consensus indicates that further refinement of this concept is required, and perhaps dedicate experimental studies to provide evidence in support of a more definitive statement in future.

Imaging remains an extremely useful tool in MSKI research but is less quantitative than some outcome measures and in particular CFU counting. Delegates showed strong support for the use of IVIS and other imaging techniques as complementary methods, but also highlighted their limitations. Use of radiographic scoring as an outcome measure was less favored, likely because of the limitations of the many scoring systems and lack of standardization of evaluations. These methods provide valuable insight into the infection status through minimally invasive methods with potential to longitudinally monitor the progress of treatment or development of infection. The consensus responses recommend these methods if available but also caution about using them as the primary evaluation. As such, they may be more reflective of boney changes secondary to infection than the infection itself.

5 | CONCLUSION

The recommendations and voting results of the 2023 ICM may be used to guide future research toward improved performance of preclinical research on treating MSKI. Overall, there was majority consensus for all the recommendations, although many responses were based on limited or low-quality evidence found in the literature and some questions had a high abstention rate. Future research should be directed toward improving the quality of evidence, particularly for questions of significance of reducing bacterial burden but not eliminating it and the refinement of polymicrobial models. Furthermore, the issue of the wide variety of infections requiring a myriad of models is an issue that may need some consolidation to improve the value of preclinical research. Understanding and overcoming these issues will improve research quality and in the long term will benefit patients and clinicians.

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